

ASSIGNMENT NO 1 – NCBI, BLAST, CLUSTAL OMEGA

Q1. SELECTING QUERY SEQUENCE:

PROTEIN

APRT - adenine phosphoribosyltransferase BETA

Homo sapiens

Summary Transcripts and proteins Orthologs Gene ontology

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 datasets

API

Gene name	adenine phosphoribosyltransferase
Symbol	APRT
Gene ID	353
Organism	Homo sapiens (human)
Gene type	protein-coding
Synonym	AMP, APRTD
Proteins	2
Transcripts	2
MANE Select	NM_000485.3
Refseq summary	Adenine phosphoribosyltransferase belongs to the purine/pyrimidine phosphoribosyltransferase family. A conserved feature of this gene is the distribution of CpG dinucleotides. This enzyme catalyzes the formation of AMP and inorganic pyrophosphate from adenine and 5-phosphoribosyl-1-

Other identifiers

Ensembl	ENSG00000198931
HGNC	HGNC:626
OMIM	102600
UniProt	P07741

Publications

mBio 2026
Differentiation-dependent
proximity proteomics identifies
novel host factors linked to HPV16
E2 function
James, Claire D, et al.

EMBO Rep 2026
Proximity-labeling proteomics
[Proximity-labeling proteomics](#)



An official website of the United States government [Here's how you know](#)



National Library of Medicine
National Center for Biotechnology Information

Protein

Protein

[Advanced](#)

FASTA

APRT [Coturnix japonica]

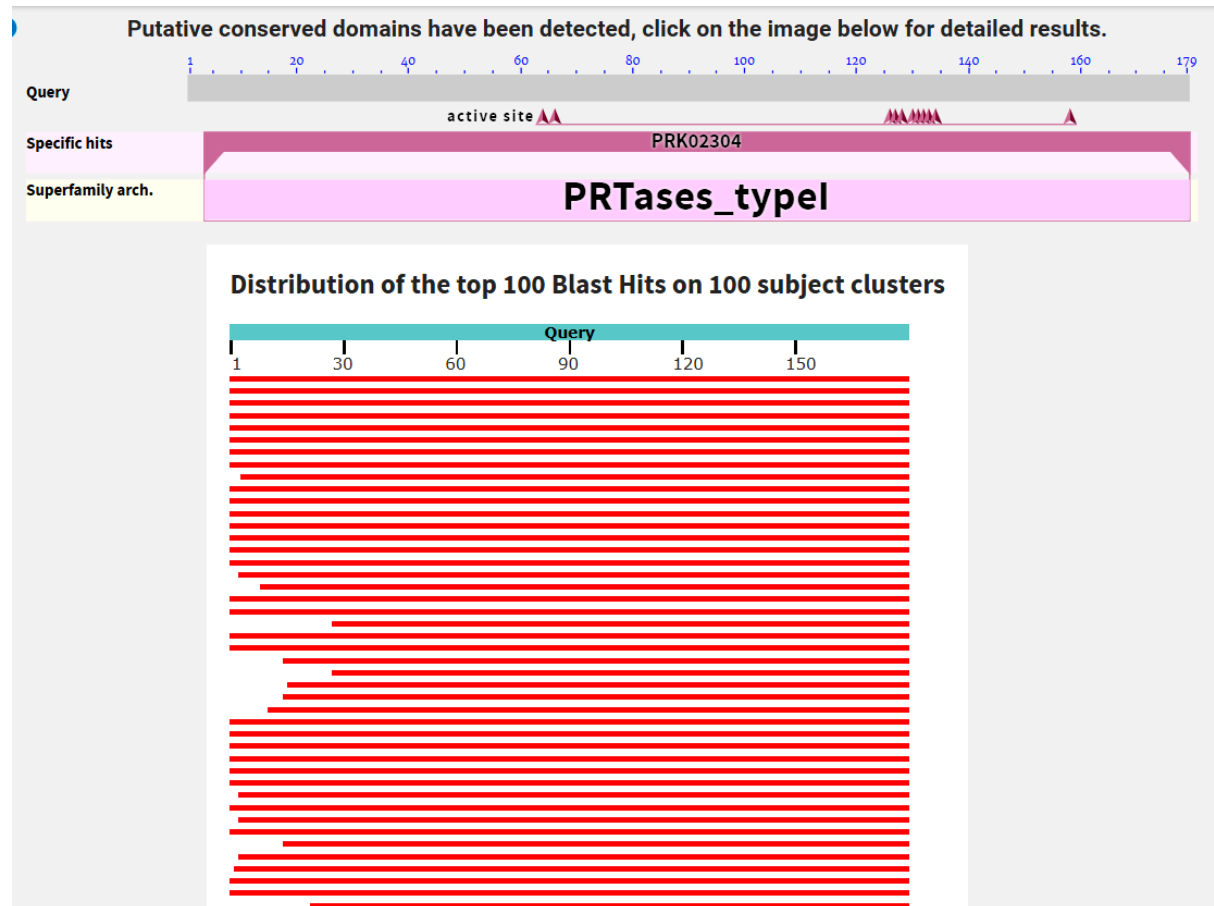
GenBank: CAP2904332.1

[GenPept](#) [Identical Proteins](#) [Graphics](#)

>CAP2904332.1 APRT [Coturnix japonica]
MSEDKLRVVRDRVRSFPDFPVPVGLFRDISPLLKDPVAFKALIDLLEDHLRASFPKIDVIAGLDSRGFLI
GPPLAQRLGVGFVLIIRKKGKLPGPTEISYSLEYGKAELEVQSDAVEAGQKVIVDDLLATGGTMRAACE
LLARLKADILECLVWIELKALRGAALKDAIPFHSLLQYD

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Q2. RUN BLAST:



Tool output

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CLUSTAL O(1.2.4) multiple sequence alignment

```
XP_036246052.1:194-372 MSDELRRLRRVDRVRFPDFPEPGVLFDRDISPLLKDPVAFRTLIDLLEDHVRASFPKIDFI 60
XP_064016188.1:1-179 MSGERLQVRDRRLRSFPDFPQGVLFDRDISPLLKDPAAFKALIDLLEDHLRASFPRIIDYI 60
XP_061216024.1:23-201 MSEEQLRRVRDRVRSPDFPVPVGRVFRDISPLLKDPAAFRTLIDLLEDHLRVSHPKIDFI 60
** *:*****: * ***** ** *****:*****:*.*.**:**

XP_036246052.1:194-372 VGLDSRGFLIGTPLAQLRGIGFVPVRKKGKLPGATESISYSLEYGKAELEVQSDAVEPGQ 120
XP_064016188.1:1-179 AGLDSRGFLIGPPLAQLRGIGFVPVRKKGKLPPTESVSYSLEYGKAELEIQSDAVEQGG 120
XP_061216024.1:23-201 AGLDSRGFLIGPPLAQLRGIGFVLIRKKGKLPPTESISYALEYGKAELEIQSDAVEAGQ 120
.***** ***** :***** ***:*****:***** **

XP_036246052.1:194-372 KVVIVDDLATGGTMRAACELLKRLKAEILECLVIELKALKGSEKLSIPFYSLLOYD 179
XP_064016188.1:1-179 KVVIVDDLATGGTMCAACELLKRLKADILECLVIELKSLRGSEKLSIPFYSLLOYN 179
XP_061216024.1:23-201 KVVIVDDLATGGTMFAACELMKRLKAEVLECLVIELKLFKGSEKLSIPFYSLLOYD 179
***** *****:*****:*****:*****:*****:*****:
```

Tool Output

Alignments

Guide Tree

Phylogenetic Tree

Results Viewers

Result Files

Submission Details

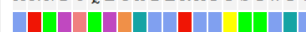
COLOR SCHEME

clustal2



LEGEND

ARNDCQEGHILKMFPSTWYVBXZ



Nightingale

3 sequences



XP_036246052.1:194-372
XP_064016188.1:1-179
XP_061216024.1:23-201

